



ADAPTIVE WIRELESS NETWORK

Wireless Networks SCE

Name	Roll no	PRN
Manas Kulkarni	311058	22311608
Samiksha Nalawade	311063	22311696
Rajlakshmi Desai	311065	22311732
Shreyash Shabadi	311068	22420082

Project Guide: Prof. Rupa Kawchale



INTRODUCTION

1

Multi-Node Network

1 AP + 14 STAs randomly placed in a 50×50 m indoor space

3

Rayleigh Fading

Multipath propagation modelled with exponential power gain

2

Adaptive Modulation

BPSK · QPSK · 16-QAM · 64-QAM — rate selected by SNR

4

Anti-Jamming

5 GHz frequency hop triggered when $\text{SNR} < 5 \text{ dB}$ under jamming



LIBRARIES USED

Library	Import Name	Purpose
NumPy	<code>numpy (np)</code>	Numerical arrays, random number generation, logarithms, spatial coordinates, Euclidean distance
Matplotlib	<code>matplotlib.pyplot (plt)</code>	Plotting network topology maps and comparative performance charts
SciPy	<code>scipy.special.erfc</code>	Computes the Complementary Error Function — required for theoretical BER in Gaussian noise channels
NetworkX	<code>networkx (nx)</code>	Graph/topology generation support (simulation primarily uses NumPy for spatial mapping)



TECHNICAL PARAMETERS

Component	Details
Protocol	IEEE 802.11 b/g/n (WiFi) at 2.4 GHz
Modulations	BPSK · QPSK · 16-QAM · 64-QAM
Channel Model	AWGN + Indoor Log-Distance + Rayleigh Fading
Key Metrics	BER · Throughput · SNR · Path Loss · Data Rate
Simulation Tool	Python (NumPy, SciPy, Matplotlib, NetworkX)
Network Size	1 AP + 14 STAs in a 50×50 m indoor space
Scenarios	Scenario A — Normal Scenario B — War-like
Bits Simulated	50,000 bits per node per scenario
Random Seed	42 — fully reproducible results

Stage 1

POINT-TO-POINT WIFI LINK (IEEE 802.11)

Standard	Modulation	Data Rate	Typical BER	Typical Throughput	Status
802.11b	BPSK	1 Mbps	~0.000000	~1.0 Mbps	EXCELLENT
802.11b	QPSK	2 Mbps	~0.000000	~2.0 Mbps	EXCELLENT
802.11g	BPSK	6 Mbps	~0.000000	~6.0 Mbps	EXCELLENT
802.11g	QPSK	12 Mbps	~0.000000	~12.0 Mbps	EXCELLENT
802.11g	16-QAM	24 Mbps	~0.000100	~24.0 Mbps	GOOD
802.11g	16-QAM	36 Mbps	~0.001200	~35.9 Mbps	GOOD
802.11n	64-QAM	48 Mbps	~0.015000	~47.3 Mbps	MARGINAL
802.11n	64-QAM	54 Mbps	~0.015000	~53.2 Mbps	MARGINAL



ADAPTIVE RATE SELECTION

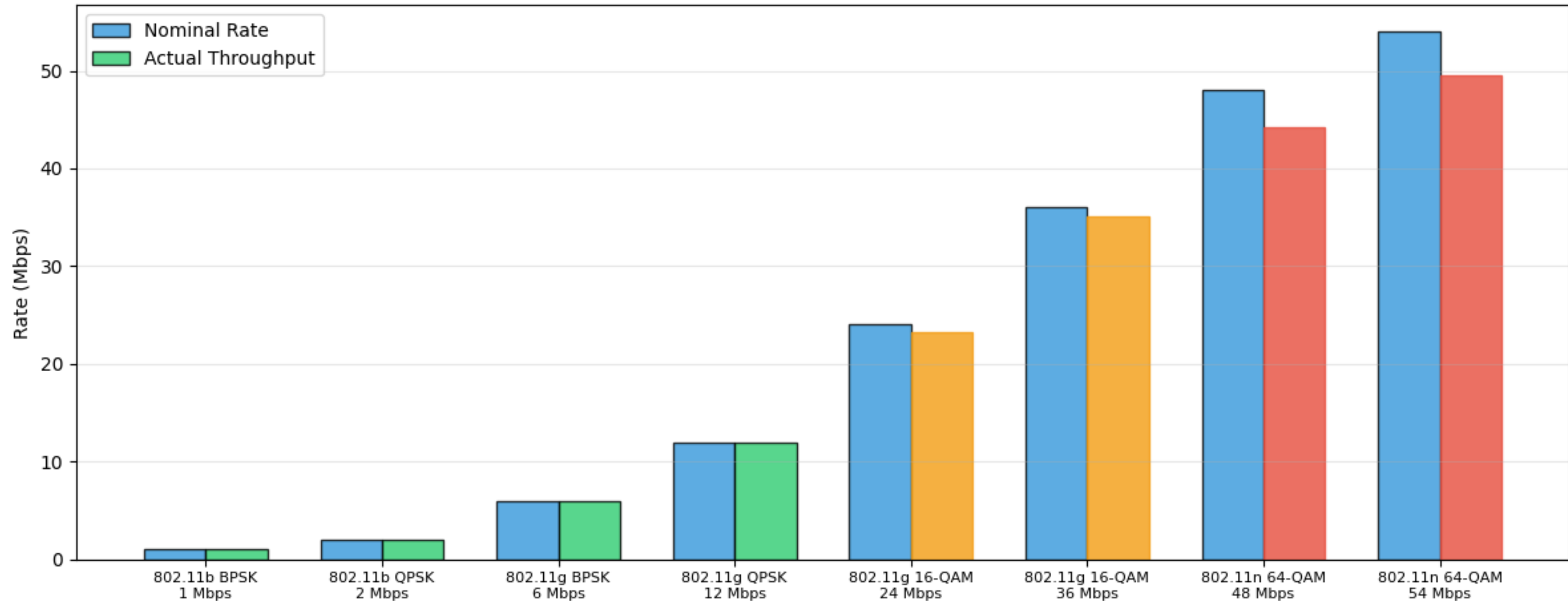
SNR Range	Selected Mode	Typical Throughput	Reason
> 17 dB	802.11n 64-QAM 54 Mbps	~51–54 Mbps	BER well below 5% — fastest mode viable
11–17 dB	802.11g 16-QAM 24 Mbps	~22–24 Mbps	64-QAM BER exceeds threshold; 16-QAM works
4–11 dB	802.11b QPSK 2 Mbps	~1.5–2 Mbps	16-QAM too noisy; only BPSK/QPSK viable
< 4 dB	No Link	0 Mbps	No modulation survives — node disconnected

SIMULATION PARAMETERS

Parameter	Value	Technical Meaning
num_bits	50,000	Total bits transmitted per node per scenario. More bits → more statistically reliable BER.
carrier_freq	2.4×10^9 Hz	WiFi 2.4 GHz band. Wavelength $\lambda = c/f = 3 \times 10^8 / 2.4 \times 10^9 = 0.125$ m.
tx_power_dBm	20 dBm	Router transmit power = 100 mW. Conversion: $P(\text{dBm}) = 10 \times \log_{10}(P_{\text{mW}})$.
max_distance	50 m	Simulation area radius. STAs beyond 50 m from AP are likely out of range.
noise_power_dBm	-77 dBm	Baseline receiver noise floor derived from thermal noise $N = kTB$.
PL_0	40 dB	Reference path loss at 1 metre from the AP at 2.4 GHz.
n_exponent_normal	3	Indoor path loss exponent for Scenario A. Quantifies signal attenuation beyond 1 m.

NOMINAL DATA RATE VS THROUGHPUT

802.11 Nominal Data Rate vs Actual Throughput at 6.0 dB SNR



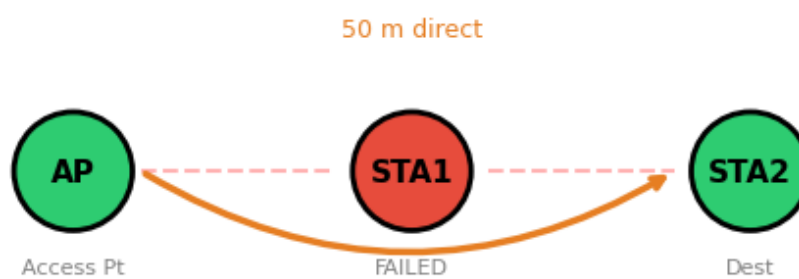
THREE FAULT SCENARIOS

S1: Normal Relay (AP->STA1->STA2)



BER=0.0 TP=24.0 Mbps

S2: Relay Failed (direct AP->STA2)

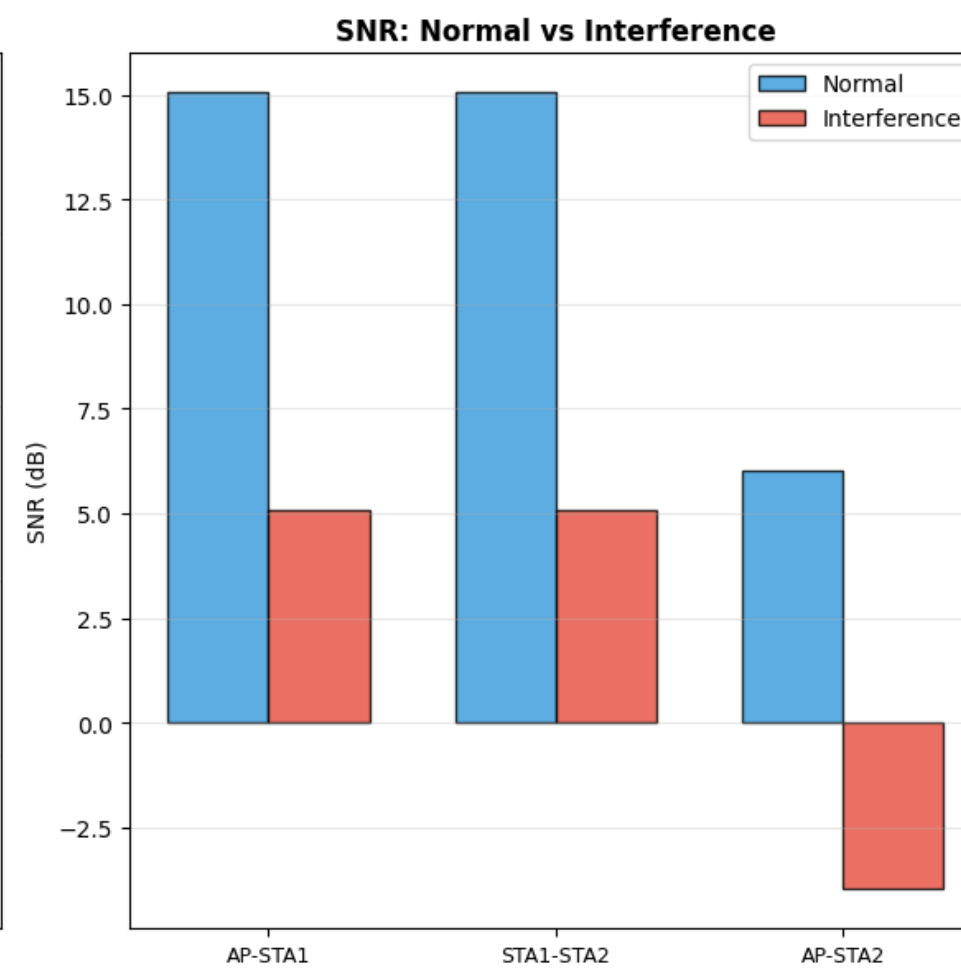
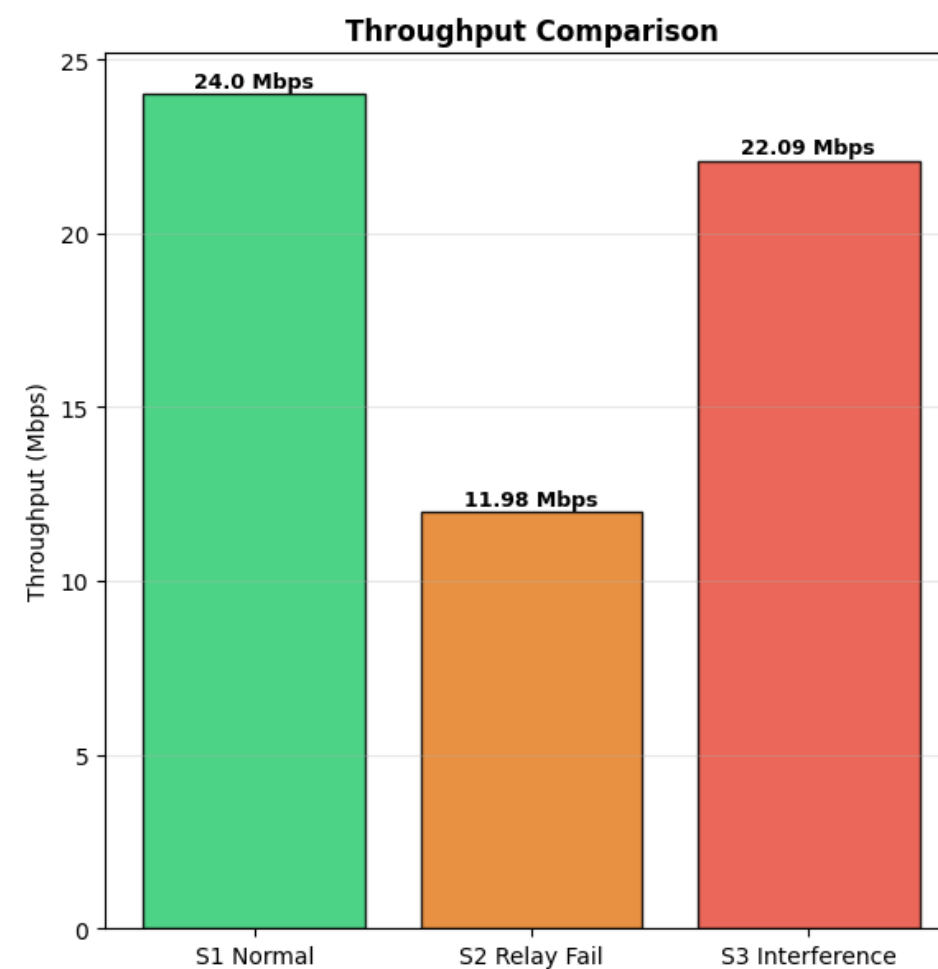
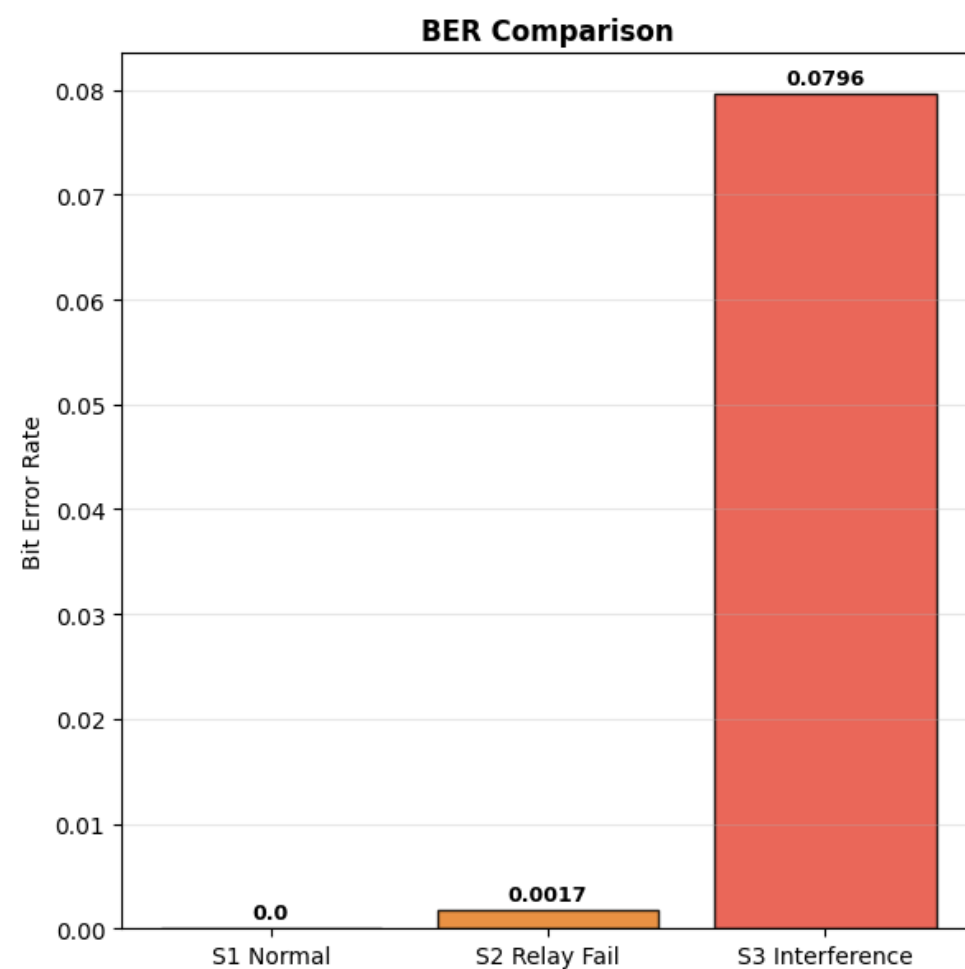


BER=0.0017 TP=11.98 Mbps

S3: High Interference (+10 dB noise)



BER=0.0796 TP=22.09 Mbps



Stage 2



CHANNEL MODELS & NOISE

Rayleigh Fading

Multipath
Propagation

$$|H|^2 \sim \text{Exp}(1)$$

Gaussian Noise

Path Loss Model

$$PL = PL_0 + 10n \cdot \log(d)$$

Interference

Jamming Scenario

$$\text{SNR}_{\text{jammed}} < 5 \text{ dB}$$

Anti-Jamming: Frequency Hopping to 5 GHz band triggered when $\text{SNR} < 5 \text{ dB}$



RAYLEIGH FADING

Property	Path Loss	Rayleigh Fading
Nature	Predictable, deterministic	Random, stochastic, instantaneous
Cause	Distance + wall absorption	Multipath reflected copies cancelling
Typical scale	Tens of dB over tens of metres	± 5 to -30 dB in milliseconds
In code	<code>get_indoor_path_loss(dist, n)</code>	<code>get_rayleigh_fading_dB()</code>
Applied in	Both scenarios always	Scenario B only (<code>apply_fading=True</code>)

Path loss is a predictable, average effect. Rayleigh fading is the unpredictable, moment-to-moment variation on top of that average.

INDOOR PATH LOSS MODEL

$$PL(d) = PL_0 + 10 * n * \log_{10}(d)$$

Symbol	Value	Meaning
PL_0	40 dB	Reference path loss at 1 m from AP (2.4 GHz)
n	3 or 4	Path loss exponent — depends on environment
d	metres	Distance from AP to STA
$\log_{10}$	—	Base-10 logarithm

Path Loss Exponent Reference Table

Environment	n Value	Typical Use
Free Space	2	Open air, line-of-sight, no obstacles
Indoor Office	3 ← Scenario A	Offices, homes, standard walls & furniture
Dense Indoor	4 ← Scenario B	Thick concrete, rubble, bombed buildings
Very Dense	5–6	Factories, underground, heavy industrial

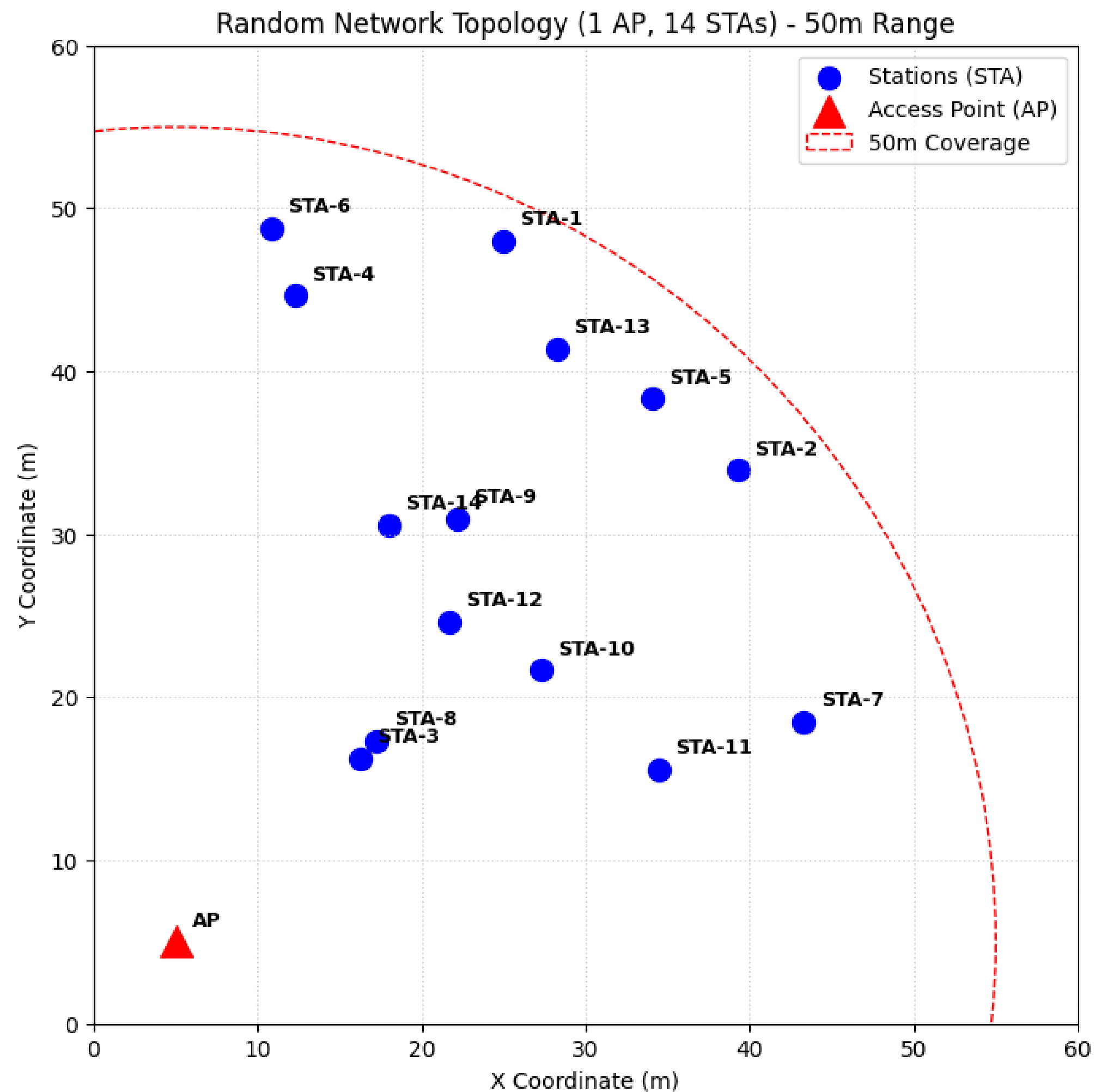
RANDOM NETWORK TOPOLOGY

15 Total Nodes

1 Access Point

14 Stations (STA)

50m Coverage Radius



SCENARIO A

NORMAL INDOOR ENVIRONMENT

Parameter	Value	Meaning
Path Loss Exp.	$n = 3$	Standard office/home. Walls reduce signal but not severely.
Extra Noise	0 dB	No interference. Noise floor stays at baseline -77 dBm.
Rayleigh Fading	OFF	No multipath fading. Channel is deterministic and predictable.
Effective Noise	-77 dBm	Baseline thermal noise only.

9.1 Expected Results by Distance

Distance	SNR Range	Mode Selected	Throughput
~0–20 m	20–30 dB	802.11n 64-QAM	~50–54 Mbps
~20–30 m	12–20 dB	16-QAM or 64-QAM	~24–50 Mbps
~30–40 m	10–18 dB	16-QAM or QPSK	2–24 Mbps
~50 m	6–10 dB	QPSK / BPSK	1–2 Mbps
> 50 m	< 4 dB	No Link	0 Mbps

SCENARIO B

WAR-LIKE / WORST CASE

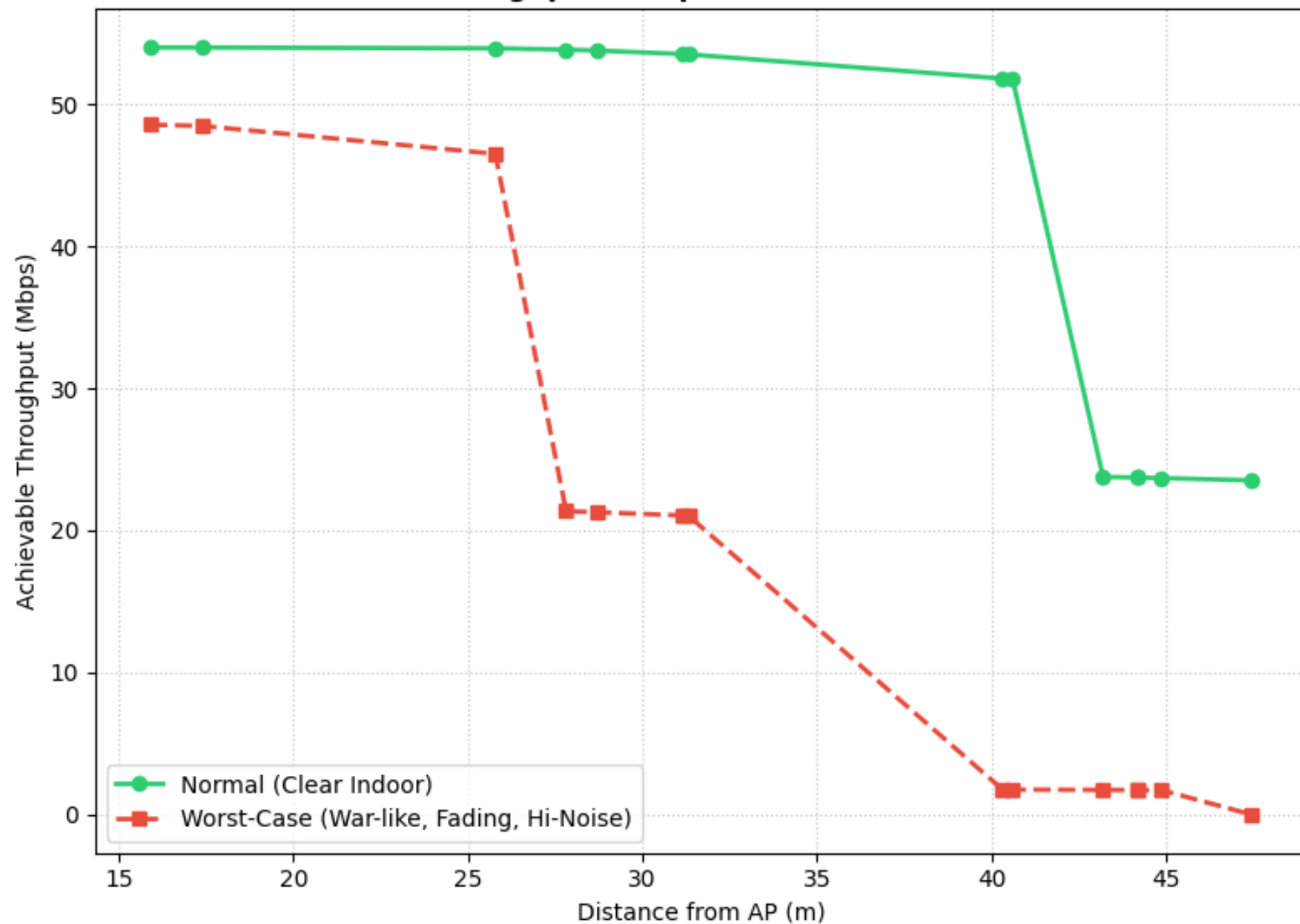
Parameter	Value	Meaning
Path Loss Exp.	$n = 4$	Dense rubble, thick concrete — signal absorbed much faster.
Jamming (extra noise)	+15 dB	Active jammer raises noise floor from -77 dBm to -62 dBm.
Rayleigh Fading	ON	Severe multipath. Every node gets a different random fade.
Effective Noise	-62 dBm	Jammer active. RX power must exceed -62 dBm for SNR > 0 dB.

10.1 Effect of Each Impairment

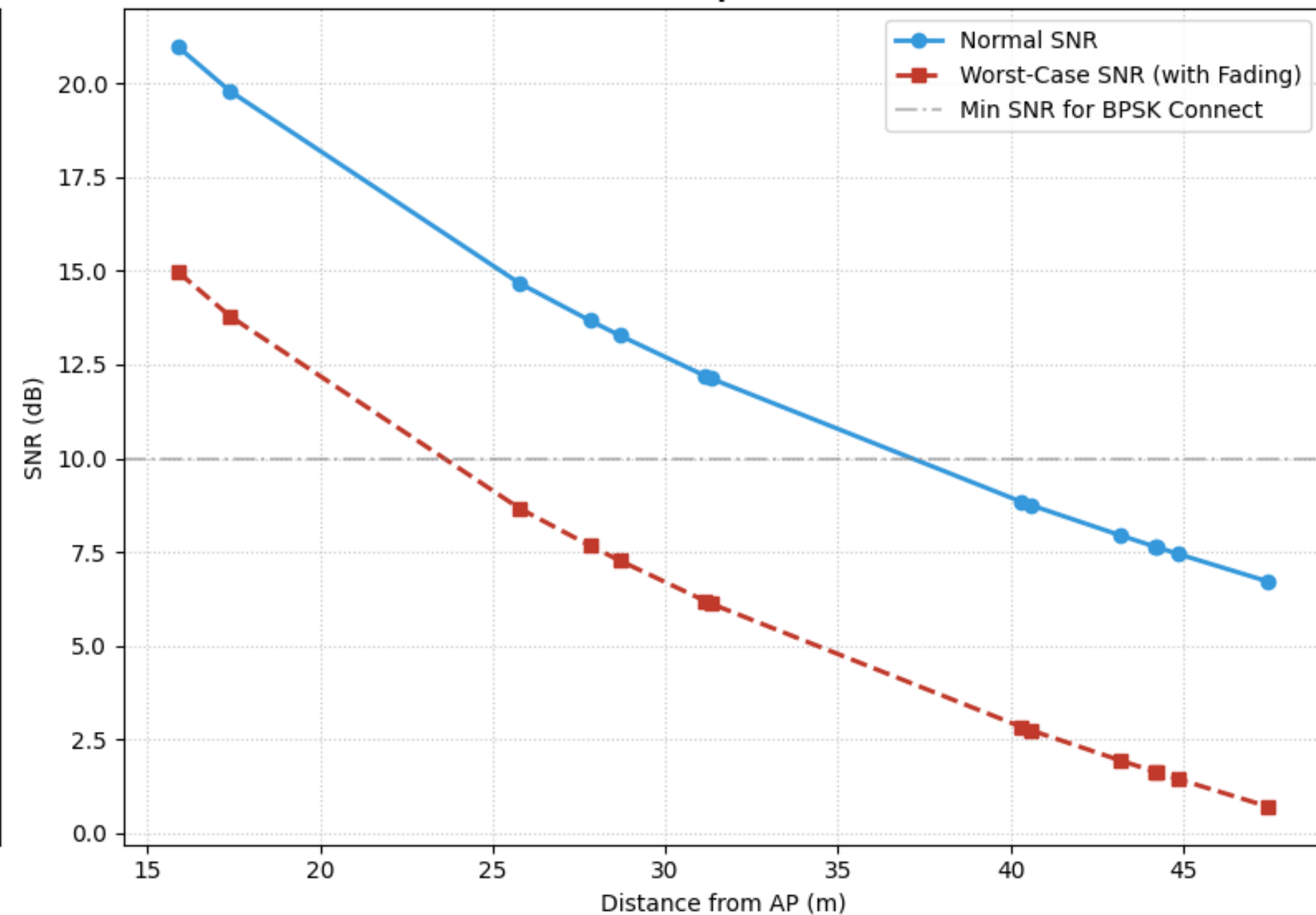
Impairment	Effect on SNR	Scope
n=4 vs n=3	Extra ~17 dB path loss at 50 m	All nodes, worsens with distance
+15 dB jammer	SNR drops by exactly 15 dB on every node	All nodes equally
Rayleigh fading	Random 0 to -30 dB drop	Each node individually, unpredictable

RESULTS- SCENARIO A VS B

Throughput Comparison vs Distance



Received SNR Comparison vs Distance



CONCLUSION / SUMMARY

Scenario A — Normal

14 / 14

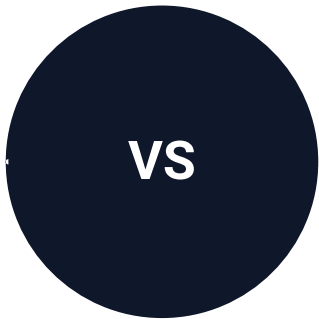
Connected Relays

100% Connectivity

42.76

Mbps

Average Throughput



Scenario B — War-Like / Worst Case

13 / 14

Connected Relays

-7.1% Drop

17.05

Mbps

Average Throughput

CONCLUSION

Severe conditions caused a 7.1% drop in node connectivity · Total network throughput capacity reduced by 60.1%

60.1%

Throughput Reduction

42.76 → 17.05 Mbps

7.1%

Connectivity Drop

1 node lost under jamming

1 / 14

Nodes Disconnected

13 of 14 nodes survived

Thank You